Spare room, shapes, and the triple-column reduction

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Abstract

This note states precisely what "spending points" means when we cut the torus, proves that it is correct, and proves the triple-column reduction used for the seam case. Every graph is finite and simple; every list has 3 colours unless stated otherwise.

1 The tool

Fix an orientation of a graph H. Its *graph polynomial* is

$$P_H(x) = \prod_{(u,v) \text{ an arc}} (x_u - x_v).$$

Known result 1 (Combinatorial Nullstellensatz, Alon [Alon])**.** *Let $P \in \mathbb{C}[x_1, \dots, x_N]$ have degree $\sum_i t_i$, with $t_i \geq 0$, and suppose the coefficient of $\prod_i x_i^{t_i}$ in P is nonzero. If $S_1, \dots, S_N \subseteq \mathbb{C}$ satisfy $|S_i| > t_i$, then there is $s \in S_1 \times \cdots \times S_N$ with $P(s) \neq 0$.*

Lemma 2. *P_H is a nonzero homogeneous polynomial of degree $|E(H)|$.*

Proof. Each factor $x_u - x_v$ is a nonzero homogeneous polynomial of degree 1, and a product of nonzero polynomials is nonzero. A product of homogeneous polynomials is homogeneous, of degree the sum of the degrees. There is one factor per edge. □

2 Spare room and shapes

Definition 3. The *spare room* of a graph H with N vertices is $d(H) = 2N - |E(H)|$. A *shape* for H is a function $\mu : V(H) \to \{0, 1, 2\}$ with $\sum_v \mu(v) = d(H)$. Its *coefficient* is

$$c_\mu = \Big[\prod_v x_v^{2-\mu(v)} \Big] P_H.$$

Lemma 4 (shape lemma)**.** *Let μ be a shape for H with $c_\mu \neq 0$. Let every vertex v have a list $L(v)$ of 3 colours, and choose any subsets $L'(v) \subseteq L(v)$ with $|L'(v)| = 3 - \mu(v)$. Then H has a proper colouring c with $c(v) \in L'(v)$ for every v.*

Proof. Put $t_v = 2 - \mu(v) \geq 0$. Then $\sum_v t_v = 2N - \sum_v \mu(v) = 2N - d(H) = |E(H)| = \deg P_H$, using Lemma 2. Choose an injective map from the set of all colours into $\mathbb{C}$ and regard each $L'(v)$ as a subset $S_v \subseteq \mathbb{C}$; then $|S_v| = 3 - \mu(v) = t_v + 1 > t_v$. By Known result 1 there is $s \in \prod_v S_v$ with $P_H(s) \neq 0$. Then $s_u \neq s_v$ for every arc (u, v), so $c(v) = s_v$ is a proper colouring with $c(v) \in L'(v)$. □

Remark 5 (pin and forbid). If $\mu(v) = 1$ we may take $L'(v) = L(v) \setminus \{\beta\}$ for any $\beta \in L(v)$: the colouring avoids β at v (*forbid*). If $\mu(v) = 2$ we may take $L'(v) = \{\alpha\}$ for any $\alpha \in L(v)$: the colouring uses α at v (*pin*). If $\mu(v) = 0$ nothing changes. Lemma 4 guarantees *some* colouring with these properties; it gives no other control, which is why the rest of a cut argument must work against every such colouring.

Corollary 6 (adjacent pins are useless)**.** *If $\mu(u) = \mu(v) = 2$ for adjacent vertices u, v, then $c_\mu = 0$.*

Proof. Suppose $c_\mu \neq 0$. Give every vertex the list $\{1, 2, 3\}$ and choose $L'(u) = L'(v) = \{1\}$. Lemma 4 gives a proper colouring with $c(u) = c(v) = 1$, which is impossible because u and v are adjacent. □

3 Counting the spare room

Lemma 7. *Let $m, n \geq 3$ and $\ell \geq 2$.*

(a) The torus $C_m \,\square\, C_n$ has spare room 0.

(b) The strip $C_m \,\square\, P_\ell$ has spare room m.

(c) Removing a 3×2 block (rows $t-1, t, t+1$, columns $j, j+1$) from $C_5 \,\square\, C_n$, $n \geq 5$, leaves a graph with spare room 5.

Proof. (a) $N = mn$ and every vertex has degree 4, so $|E| = 4N/2 = 2N$.

(b) $N = m\ell$. Each of the ℓ columns is a copy of C_m, giving $m\ell$ edges. Between consecutive columns there are m row edges, and there are $\ell - 1$ such pairs, giving $m(\ell - 1)$ edges. So $|E| = 2m\ell - m = 2N - m$.

(c) The torus has $5n$ vertices and $10n$ edges. The block has 6 vertices, each of degree 4 in the torus, so the sum of their degrees is 24. The block contains 7 edges (two vertical edges in each of its 2 columns, and 3 horizontal edges), each counted twice in that sum. So $24 - 14 = 10$ edges leave the block. Removing the block deletes its $7 + 10 = 17$ incident edges. The remainder has $N = 5n - 6$ vertices and $10n - 17 = 2(5n - 6) - 5$ edges. □

Remark 8. Since $n \geq 5$, the 10 edges leaving the block go to 10 distinct vertices (the *border*), and each border vertex has exactly one neighbour in the block. The 5 units of spare room are spent on the border, with shapes whose coefficients are computed exactly (`ws-W14/hole_coeff.py`; 1,256 placements nonzero for every odd $n \geq 5$, proved with `ws-W14/hole_prove.py`).

4 The triple-column reduction

Throughout this section L is a list assignment of $C_5 \,\square\, C_n$ with $n \geq 5$ odd. A *column pattern* for column j is a proper colouring P of the 5-cycle (rows $\mathbb{Z}_5$) with $P_r \in L(r, j)$. Two patterns P, Q are *compatible*, written $P \sim Q$, if $P_r \neq Q_r$ for every row r. A colouring of the torus is the same as one pattern per column with consecutive patterns compatible.

Known result 9 (cycles; Erdős, Rubin and Taylor [ERT])**.** *An odd cycle whose lists all have size at least 2 can be coloured unless all its lists are the same 2-set.*

Lemma 10 (degenerate block)**.** *Let $E = \{e_1, e_2\}$, and let $x = (x_0, \dots, x_4)$ be a proper colouring of C_5 with no $x_r \in E$. Let $A_r = E \cup \{x_r\}$. Let U, V be any proper colourings of C_5 (with any colours) such that $U_r \neq x_r$ and $V_r \neq x_r$ for all r. Then there are proper colourings c^1, c^2, c^3 of C_5 with $c^k_r \in A_r$ and $U \sim c^1 \sim c^2 \sim c^3 \sim V$.*

Proof. Whether $U \sim c$ holds for c with $c_r \in A_r$ depends only on whether U_r equals e_1, equals e_2, or lies outside A_r (the case $U_r = x_r$ is excluded), and on nothing else about the colours; the same holds for V. So it suffices to check every pair of type vectors in $\{e_1, e_2, \text{other}\}^5$ for which U and V are proper (no two cyclically adjacent rows both e_1 or both e_2), with patterns c^k ranging over the 81 proper colourings of C_5 from the lists A_r. All 6,561 pairs admit such a path (`ws-W17/degenerate_fold.py`). □

Theorem 11 (triple-column reduction)**.** *Suppose $C_5 \,\square\, C_{n-2}$ is 3-choosable. If three consecutive columns $j, j+1, j+2$ of $C_5 \,\square\, C_n$ have identical lists, $L(r, j) = L(r, j+1) = L(r, j+2) =: A_r$ for all r, then $C_5 \,\square\, C_n$ is L-colourable.*

Proof. Delete columns $j+1$ and $j+2$ and join column j to column $j+3$ by the row edges $(r, j)(r, j+3)$. The result is a copy of $C_5 \,\square\, C_{n-2}$ with the inherited lists; by assumption it has a colouring φ. Let P, U, V be the patterns of φ on columns j, $j-1$, $j+3$. Since φ is proper, $U \sim P$ and $V \sim P$. We keep φ outside columns $j, j+1, j+2$ and choose new patterns c^1, c^2, c^3 for them, from the lists A_r, with $U \sim c^1 \sim c^2 \sim c^3 \sim V$. This gives a proper colouring, because every edge of $C_5 \,\square\, C_n$ outside these three columns is an edge of the smaller torus.

Case 1: the lists $A_r \setminus \{P_r\}$ (each of size 2) are not all the same 2-set. By Known result 9 there is a proper colouring W of C_5 with $W_r \in A_r \setminus \{P_r\}$, so $W \sim P$. Take $c^1 = P$, $c^2 = W$, $c^3 = P$. Then $U \sim P \sim W \sim P \sim V$.

Case 2: $A_r \setminus \{P_r\} = E$ for all r, for one 2-set E. Then $A_r = E \cup \{P_r\}$ with $P_r \notin E$, and P is proper. This is Lemma 10 with $x = P$; it supplies c^1, c^2, c^3. □

Remark 12. Theorem 11 turns the main theorem into an induction on odd n, with base case $n = 3$, which is Theorem A ($C_5 \,\square\, C_3 \cong C_3 \,\square\, C_5$). Every block of a piecewise row-constant assignment has at least three identical consecutive columns, so Theorem 11 settles that case with no condition on overlaps.

References

[Alon] N. Alon, Combinatorial Nullstellensatz, *Combin. Probab. Comput.* 8 (1999), 7-29.

[ERT] P. Erdős, A. L. Rubin and H. Taylor, Choosability in graphs, *Congr. Numer.* 26 (1979), 125-157.